CIS-445 Machine Learning

Group Homework #5

Group # and Names of group members: **Group #5 Dylan Shea, Channel Caldwell, Clay Peterson**
Worth 50 points Due date: See Backboard

Logistics

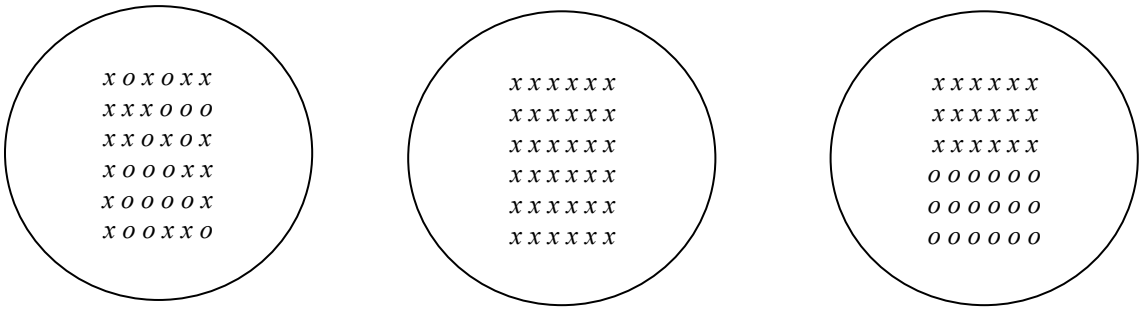
1. Get in touch with your group. (See Groups folder on Blackboard.)
2. Discuss and work all 3 problems collectively with your group via E-mail, Discussion Forum, Blackboard Collaborate Ultra, and/or MS Teams. (Do not divide the work among group members.)
3. Choose a recorder to prepare the final copy (one per group) and submit it via the Blackboard Assignments/Homeworks folder.
4. Be sure all group members' names are on final copy. Do **not** add names of your group members who did not participate in the assignment or whose contribution was minimal.

Objectives: Decision Trees, Classification Task, Entropy (the measure of disorder/impurity in a cluster of data), Entropy Gain

You must show your calculations!

Problem 1 (10 points)

There are three clusters: A, B, and C. The *o*'s and *x*'s in the clusters indicate two categories "yes" and "no", respectively. For example, an *o* and *x* may indicate that a customer is a widget buyer and non-widget buyer, respectively. Compute the disorder (*Entropy*) for each cluster, where p_i is the probability of the ith message/category occurring in a particular cluster.



A B C

In this example, *i*=2.

$$Entropy = -\sum_i p_i \log_2 p_i$$

Entropy(A): o's= 17 and x's=19 po=17/36=.472 px=19/36=.528
Entropy(Ao)= -(.472log2.472)=-(.427*-1.228)=-(-.524) = .511
Entropy(Ax)=-(.528log2.528)=-(.528*-.921)=-(-.486)= .486
Entropy(A)= .511 + .486=.997

Entropy(B): o's= 0 and x's=36 po=0/36= 0 px=36/36=1
Entropy(Bo)= -(0log20)=0

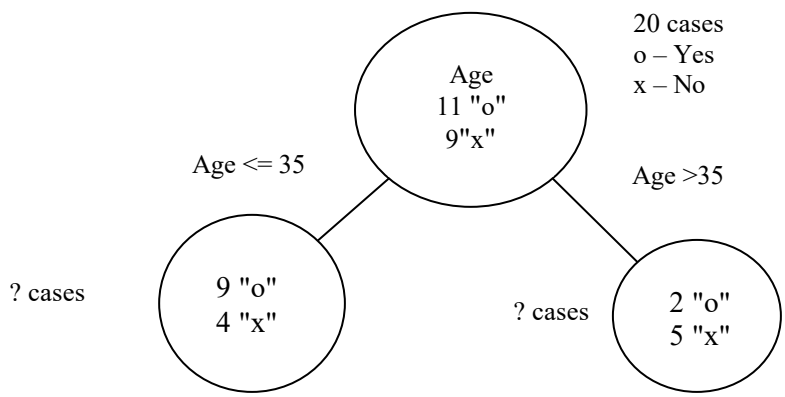
Entropy(Bx) = -(11og2*1)= -(1*0) = 0
Entropy(B)=0

Entropy(C): o's= 18 and x's=18 po=18/36=.5 px=18/36=.5
Entropy(Co) = -(.5log2.5)= -(.5*-1)= -(-.5)= .5
Entropy(Cx) = -(.5log2.5)= -(.5*-1)= -(-.5)= .5
Entropy(C)= .5+.5= 1

Entropy for cluster A: .997
Entropy for cluster B: 0
Entropy for cluster C:1

Which cluster is the purest and what is its entropy? Cluster B and entropy is 0
Which cluster is the most impure and what is its entropy? Cluster C and entropy 1

Refer to Figures 10.1 and 10.5 on pp. 147 and 150, respectively, in your CoursePack posted in the *Course Documents/Week 7* folder on BB. The ID3 and C4.5 algorithms place each of the 6 variables at the top of the decision tree, and for each variable the algorithms determine the split that will maximize the information gain. In other words, from many possible splits on the 6 variables, the algorithms will choose the split that produces purer clusters than other splits. Let us assume that the variable Age is placed at the top of the tree and the split at the value of Age = 35. Calculate the gain in purity on variable Age resulting from this split only, which results in two branches on Age≤35 and Age>35. Compare it to the gain=0.28 that resulted from placing the variable Residence at the top of the tree and splitting it on "NY" (New Yorkers) and "≠NY" (Non New Yorkers) as shown in Figure 10.5. Let us verify if the authors of the text indeed calculated all possible splits for all the variables, and the value 0.28 is the highest gain one can obtain for this data. Replace the question marks "?" with the number of cases "o" and "x" that fall into the clusters.



After counting the total number of 20 points

Yes(o) = 11, No (X) = 9

Less than or equal to 35:
Yes(O) = 9, No(X) = 4

Greater than 35

Yes = 2, No = 5

Entropy Before split = -0.55logsub2(0.55) – 0.45logsub2(0.45)

= 0.993

Entropy Before Split – **Younger than 35**
9/13 = 0.692, 4/13 = 0.308
H1 = -((0.692logsub2 0.692) – (0.308logsub2 0.308))
= ((0.692 * -0.53) – (0.308 * -1.70))
= 0.88

Older than 35 -> 2/7 = .286, 5/7 = .714
H2 = -(0.286logsub2 0.286) – (0.714logsub2 0.714)
= -((0.286 * -1.81) – (.714 * -0.48)
= 0.86

Weighted:

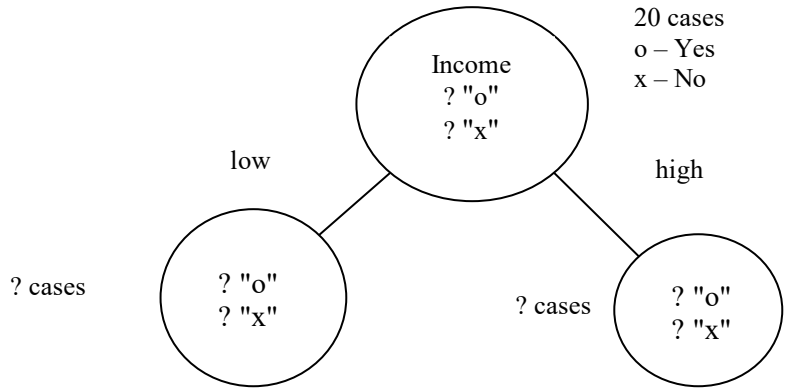
13/20(.881) + 7/20(.863) = HSplit = 0.875

Gain(Age) = 0.993 – 0.875 = 0.188

Gain(Age) = 0.118
Gain(Residence) = 0.28

Problem 3 (20 points)

The class demo has already shown that the largest information gain is obtained from the decision tree for variable Income (with the split on "high" and "low"). Compute the gain resulting from this split. Compare this gain to the gain calculated in Problem 2 above and the gain shown in Figure 10.5, p. 150 that we also calculated in class (video recording). What split and on what variable generates the largest gain? Consider variable Income, variable Age with the split at 35, and variable Residence with the split on "NY" and "≠NY". Replace the question marks "?" with the number of cases "o" and "x" that fall into the clusters.



Total income cases: 20

- "o" = 11
- "x" = 9

Low Income cases: 9

- "o" = 8
- "x" = 1

High Income Cases: 11

- "o" = 3
- "x" = 8

Entropy Before Split
=-((0.45*-1.152+0.55*-0.8625)+(0.55*-0.8625-0.45*-1.152))
0.949

Entropy Low Income
=-((0.889*-0.16993+0.111*-3.16993)+(0.111*-3.1699+0.889*-0.16993))
1.006

Entropy High Income
=-((0.727*-0.45943+0.2727*-1.87447)+(0.2727*-1.87447+0.7272*-0.45943))
1.690

Weighted
=(9/20)*(1.006)+(11/20)*(1.69)
1.3822
Gain(Income)
=1.986-1.3822
0.6038

Gain(Income)=0.6038
Gain(Age) = 0.118

Gain(Residence) = 0.28

The gain in income is the largest of the three.